TIAGo Task 1
Run & Record Guide

Autonomous frontier exploration and SLAM mapping in Gazebo

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This guide records the server commands and display setup used in our session. Run all ROS commands on the Linux server, not in the Mac terminal unless that terminal is already connected by SSH.

1. Confirm the server and desktop

Connect to the existing remote desktop and open a terminal inside it. An SSH terminal also works if it has the correct desktop display and permissions. Keep the remote desktop visible for Gazebo and RViz.

```
hostname
echo "DISPLAY=$DISPLAY"
command -v vglrun
pgrep -af 'gzserver|gzclient'
```

Observed working setup: hostname **Mohammad-CUDA-vnet184**; desktop display **:1.0**; VirtualGL at **/usr/bin/vglrun**; NVIDIA L40 GPU. If Gazebo is already running, inspect that instance before starting another one.

2. Confirm GPU rendering

```
vglrun -d egl glxinfo -B
```

Our successful test reported **direct rendering: Yes**, NVIDIA Corporation, NVIDIA L40, and OpenGL 4.6. The **-d egl** option is essential for this server configuration: ordinary vglrun tried the unavailable display :0.

3. Terminal 1 - start Gazebo and TIAGo

```
source /opt/ros/humble/setup.bash
source /home/ml/tiago_exam_2025_ws/install/setup.bash
export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp

vglrun -d egl ros2 launch tiago_exam tiago_exam.launch.py \
  world_name:=example \
  moveit:=false
```

Checkpoint: Gazebo displays the example world and TIAGo. Wait for the robot to spawn and controllers to initialize. Keep Terminal 1 running. Do not reset the world or restart Gazebo during mapping.

Start navigation & SLAM

4. Terminal 2 - launch Nav2, SLAM and RViz

Open a second server terminal and source the same environment:

```
source /opt/ros/humble/setup.bash
source /home/ml/tiago_exam_2025_ws/install/setup.bash
export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp

vglrun -d egl ros2 launch tiago_2dnav \
  tiago_nav_bringup.launch.py \
  is_public_sim:=True \
  world_name:=example \
  slam:=True
```

These are the argument names and capitalization confirmed by the installed launch file. This launch started SLAM Toolbox, Nav2, the map saver, and RViz. Keep Terminal 2 running.

5. Check startup before moving

In our log, Nav2 reached **Managed nodes are active**, SLAM Toolbox registered the lidar, and RViz received an occupancy map. A partial map can appear while the robot is stationary because the laser already sees nearby free space and obstacles.

The initial costmap warnings stopped in the supplied log after **StaticLayer: Resizing costmap**. If out-of-bounds or transform errors continue, resolve them before starting exploration. A process-started message alone does not confirm readiness.

Optional readiness checks - a separate terminal

These are additional diagnostic checks, not commands whose output was captured in our session. Source the environment first in any new terminal.

```
ros2 lifecycle get /bt_navigator
ros2 lifecycle get /planner_server
ros2 lifecycle get /controller_server
ros2 service list | grep /map_saver/save_map
ros2 topic list | grep -E '^/(map|scan|clock)$'
```

The navigation lifecycle nodes should be active; the map-saving service should be present. A listed topic does not by itself prove fresh messages are arriving.

For a fresh mapping demonstration

Use a fresh SLAM session rather than loading a saved map. Do not launch the frontier explorer until recording is ready. If repeating after a completed run, stop the old explorer, stop Terminal 2, then Terminal 1; relaunch Terminal 1 followed by Terminal 2. This discards the current unsaved SLAM session, so save any result you need first.

Arrange the recording view

6. Position Gazebo and RViz

Place **Gazebo on the left** and **RViz on the right**, without overlap. Keep the whole world visible in Gazebo. In RViz, zoom out enough to leave room for the map to expand.

Open **Panels > Displays**, or use the small triangle along the left edge of RViz to reveal the hidden Displays dock.

7. RViz settings used for a clear map

• **Global Options:** Fixed Frame = map. Check that Global Status is OK.

• **Enable:** Map, LaserScan, RobotModel, Global Planner > Path, and Controller > Local Plan.

• **Disable:** TF and Amcl Particle Swarm.

• **Global Planner:** disable Global Costmap, Downsampled Costmap and VoxelGrid. Keep Path enabled.

• **Controller:** disable Local Costmap, Trajectories and VoxelGrid. Polygon may remain enabled to show the footprint.

• **Optional cleanup:** disable Bumper Hit and other unused diagnostic displays. Hide the Displays dock once configured.

These checkboxes change only RViz visualization; they do not disable Nav2 obstacle avoidance or its internal costmaps.

The screenshots showed RobotModel initially unchecked. Enable it; if the model does not appear, expand it and inspect its Status rather than assuming it is working.

8. Understand what viewers see

White: known free space. **Black:** occupied cells/obstacles. **Grey-green:** unknown map cells with the colour scheme used in our session. The space outside the map is RViz background. A path appears after a navigation goal is selected; no path before starting exploration is expected.

9. Start recording before exploration

Mac screen recording: press Shift + Command + 5, choose Record Selected Portion, select both remote application windows, and click Record. Stop with the menu-bar stop button. You can also film in landscape with an iPhone held steady, but direct screen capture gives sharper text and maps.

Capture about five seconds of the initial partial map. Keep recording through exploration and leave the final map visible for several seconds. Capture the terminal completion and save messages separately if needed.

Run the frontier explorer

10. Terminal 3 - create a unique map destination

Open a third server terminal. The installed executable was confirmed as **tiago_exam frontier_explorer.py**, and its map-saving parameter is **map_url**. Provide a path stem without .pgm or .yaml.

```
source /opt/ros/humble/setup.bash
source /home/ml/tiago_exam_2025_ws/install/setup.bash
export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp

task1_map="/home/ml/tiago_exam_2025_ws/maps/"
task1_map="${task1_map}task1_linkedin_$(date +%Y%m%d_%H%M%S)"

echo "Map destination: $task1_map"

ros2 run tiago_exam frontier_explorer.py --ros-args \
  -p use_sim_time:=true \
  -p map_url:="$task1_map"
```

Each run gets a timestamped basename, preserving earlier map results. Keep this terminal open and avoid launching a second explorer or sending manual RViz goals while it runs.

11. Let exploration finish automatically

The explorer selects frontier navigation goals while SLAM builds the map and Nav2 moves the robot. Watch the map grow in RViz and the robot move in Gazebo. Do not stop the node just because the robot briefly pauses between goals.

Look for these message markers:

```
EXPLORATION COMPLETE
MAP SAVE SUCCESS
```

The earlier validated run reported completion after three consecutive rounds with no useful reachable unexplored frontier. The installed script requests automatic saving through **/map_saver/save_map**. Keep SLAM and the map saver running until saving succeeds.

12. Verify the saved files

If the explorer returns to the shell prompt, run the following in Terminal 3. If it remains running after reporting success, use another sourced terminal and copy the exact printed destination into task1_map.

```
ls -lh "${task1_map}.pgm" "${task1_map}.yaml"
cat "${task1_map}.yaml"
```

Both files should exist and be non-empty. The YAML contains the image reference, map resolution, origin and thresholds. Keep the YAML and its referenced PGM together.

Evidence note: the recording supplied in our session shows map growth and navigation. It does not show terminal confirmation of completion or saving. Verify those messages/files for the current run before describing map-saving success.

Troubleshooting & shutdown

Graphics error: BadAlloc / GLXMakeCurrent

Launching Gazebo directly produced a GLX graphics error in both gzserver and gzclient. The command that worked used **vglrun -d egl**. Use it for both the simulation and navigation launch, which also starts RViz.

```
vglrun -d egl glxinfo -B
```

VirtualGL: Could not open display :0

Plain vglrun tried :0, but only the X1 socket existed. Use the tested EGL option; do not change DISPLAY to :0. If the EGL test fails now, inspect the environment before launching again:

```
echo "DISPLAY=$DISPLAY"
ls -l /tmp/.X11-unix/
nvidia-smi -L
printenv VGL_DISPLAY
```

Controllers waiting / spawn service unavailable

In the failed launches, controller waiting messages followed the Gazebo graphics failure. Fix the first graphics or Gazebo error before troubleshooting each controller separately. After a failed attempt, stop its launch with Ctrl+C and allow shutdown to finish before retrying.

Map or robot missing in RViz

Confirm Fixed Frame = map, Map and LaserScan are checked, and RobotModel is enabled. Expand a display to read its Status. Persistent timestamp or transform errors require investigation; do not reset Gazebo time while Nav2 and SLAM remain running.

Automatic map saving fails

Keep Terminal 2 running and confirm /map_saver/save_map exists. Check the printed map destination, folder permissions and the explorer error text. If needed, this additional manual fallback saves the current /map under the same unique stem; use only after confirming the automatic files were not saved:

```
ros2 run nav2_map_server map_saver_cli \
  -f "$task1_map" --ros-args -p use_sim_time:=true
```

This fallback was not exercised in our session. Verify both files afterward. Manual saving does not prove that the explorer completed automatically.

13. Shut down in order

After confirming the files and stopping the recording: stop the explorer in Terminal 3 with Ctrl+C if still running, then stop navigation/SLAM in Terminal 2, then Gazebo in Terminal 1. Allow each launch to finish shutting down.

Working launch reminder: **Gazebo → Nav2 + SLAM + RViz → recording → explorer → verify save → stop recording → shutdown.**